EXPERIMENT NO: 1

DATE:

PREPARATION OF FERROUS AMMONIUM SULPHATE (MOHR'S SALT)

AIM : To prepare ferrous ammonium sulphate crystals,

APPARATUS: china dish, wire gauze ,

CHEMICALS REQUIRED : Ferrous sulphate ,Ammonium sulphate and dilute sulphuric acid

THEORY : Mohr salt is prepared by mixing equimolar mixture of hydrated ferrous sulphate

and ammonium sulphate in water containing small amounts of dil. Sulphuric acid

$$FeSO_4.7H_2O + (NH_4)_2SO_4 \longrightarrow FeSO_4.(NH_4)_2SO_4.6H_2O + H_2O$$

PROCEDURE : Take 250 ml beaker and add 7g of $FeSO_4$ and 3.5 g of $(NH_4)_2SO_4$.Add 2 to 3 ml of dilute sulphuric acid to prevent hydrolysis of $FeSO_4$.
In another beaker boil about 25 ml of water and add the boiling hot water to the contents of the first beaker.stir it well with a glass rod until the salt is completely dissolved.Transfer the solution into the china dish and concentrate it to crystallization point.Place the china dish containing solution in a beaker full of cold water.On cooling crystals of mohr's salt separates out.

RESULT : Pale green crystals of Mohr salt is obtained.

Mass of Mohr salt = g

EXT NO : 2

Date:

PREPARATION OF STARCH SOL

AIM

To prepare a lyophilic sol of starch

MATERIALS REQUIRED

- Soluble starch
- Distilled water
- 250 ml beaker
- 50 ml beaker
- Glass rod
- Funnel
- Filter-paper
- Tripod stand
- Wire-gauze
- Bunsen burner

THEORY

Lyophilic means ‘liquid-loving’ or ‘solvent- attracting’. This means that in this colloidal solution there is a strong attraction between the dispersed phase and dispersion medium. Starch forms lyophilic sol when water is used as the dispersion medium. The formation of sol is accelerated by heating. Starch sol can be prepared by heating it and water at 100 °C. It is quite stable and is not affected by the presence of any electrolytic impurity.

PROCEDURE

Weigh a small quantity (1 g) of soluble starch on an electronic balance.Transfer the weighed quantity of starch into a watch glass and add few drops of distilled water to make a paste. Take about 100 ml of distilled water in a 250 ml beaker and heat the beaker till the water starts to boil. Pour the starch paste slowly into the boiling water while stirring using a glass rod. Continue boiling for about 3 minutes and then allow the beaker to cool.

PRECAUTIONS

- The apparatus used for preparing the sol should be properly cleaned.
- Distilled water should be used for preparing sols in water.
- Starch should be converted into a fine paste before adding to boiling water.

- Starch paste should be added in a thin stream to boiling water.
- Constant stirring of the contents is necessary during the preparation of the sol.

RESULT

Starch sol is prepared.

EXPERIMENT NO : 3

Date:

PREPARATION OF FERRIC HYDROXIDE SOL

AIM

To prepare a lyophobic sol of ferric hydroxide

MATERIALS REQUIRED

- 2% solution of ferric chloride
- Distilled water
- 250 ml beaker
- Dropper
- Wire gauze
- Tripod stand
- Bunsen burner

THEORY

Lyophobic means 'liquid-hating'. That means in these sols, there is little or no interaction between the dispersed phase and the dispersion .These sols are easily precipitated by the addition of small amounts of electrolyte, by heating or by shaking, therefore these sols are relatively less stable than lyophilic sols. They need stabilising agents for their preparation. Examples of lyophobic sols include sols of metals and their insoluble compounds like sulphides and oxides.

Ferric hydroxide forms lyophobic sols on treatment with water. Ferric hydroxide sol is prepared by the hydrolysis of ferric chloride with boiling distilled water. The reaction takes place is as follows.

$$\underset{\text{Ferric chloride}}{FeCl_3(aq)} + 3H_2O(l) \xrightarrow{\text{Boil}} \underset{\text{Red sol}}{Fe(OH)_3(S)} + 3HCl(aq)$$

PROCEDURE

Take 50 mL of water in a beaker. Boil the water and add ferric chloride solution dropwise to the boiling water using a dropper. Continue heating until a deep red or brown solution of ferric hydroxide is obtained. Keep the contents of the beaker undisturbed for some time at room temperature

PRECAUTIONS

- Add ferric chloride solution dropwise.
- Heating is continued till the desired sol is obtained.
- Hydrochloric acid formed as a result of hydrolysis of ferric chloride is removed by dialysis process otherwise it would destabilise the sol.

RESULT

Ferric hydroxide sol is prepared.

EXPERIMENT NO : 4

Date:

TEST FOR PROTEINS, FATS AND CARBOHYDRATES

EXPERIMENT	OBSERVATION	INFERENCE
Test for Proteins To a little of the given milk, 2ml of NaOH solution is added followed by 1% $CuSO_4$ solution. **Xanthoproteic Test:-** To a little of the given milk, conc. Nitric acid is added and heated.	Bluish violet colourisation. Yellow precipitate	Presence of Proteins. Presence of Proteins.
Test for Fats **Solubility Test:-** The solubility of given substance in water and chloroform is noted. **Translucent spot test:-** Press a little of the substance in the folds of the filter paper. To a few drops of phenolphthalein, borax is added followed by small amount of given oil.	Insoluble in water but soluble in chloroform. Translucent or greasy spot appears on unfolded paper. Pink colour disappears.	Presence of Fats. Presence of Fats. Presence of Fats.

Test for Carbohydrates

A little of the given substance is dissolved in water. It is then treated with 1 mL of Molisch's reagent followed by a few drops of conc. sulphuric acid along the sides of the test tube.	Violet colouration at the junction of the two liquids.	Presence of carbohydrates.
Tollen's Test: To little of the given substance add Tollen's reagent and heat in a water bath.	Appearance of silver lining along the sides of the test tube.	Presence of reducing sugar.
Fehling's solution test:- To a little of the given substance, fehling's solution 'A' and 'B' are added in equal amounts and heated in a water bath.	Reddish precipitate is obtained.	Presence of reducing sugar.
Test for Starch To a little of the given solution, add iodine solution.	Blue black colour.	Presence of Starch.

EXPERIMENT No: 5

Date:

PAPER CHROMATOGRAPHY

AIM

To separate constituents present in a mixture containing lead and cadmium ions.

THEORY

Paper chromatography is a type of chromatography in which a special adsorbent paper is used instead. Moisture adsorbed by this acts as a stationary phase and the solvent as a moving phase. The mixture to be separated or analyzed is put at one end of the paper strip as a small spot. The solvent moves up the paper due to capillary action and the components of the mixture rise up at different levels and thus get separated from one another.

Lead and cadmium ions will move up at different rates and therefore get separated. Since the ions are colorless, this strip is sprayed with sodium sulphide solution so that lead ions form a black spot and cadmium ions form a yellow spot. Mobile phase will be a dilute solution of HCl.

PROCEDURE

- Make a spot of mixture of soln. of lead nitrate and cadmium mixture in the center of the reference line and dry the spot.
- Add dil. Soln. of HCl in a glass jar, and suspend chromatographic paper strip with the help of a glass rod in such a way that the spot does not directly dip into the solvent.
- Keep the jar undisturbed and allow the mobile phase to move 10-12 cm, above the pencil line.
- Remove the paper strip and mark the solvent front with the pencil. ● Dry the strip and then spray sodium sulphide solution over it. ● Identify the ion from its color and calculate Rf Value.

RESULT

The R_f value of Pb^{2+} ion (black spot) is =
The R_f value of Cd^{2+} ion (yellow spot) is =

OBSERVATIONS

SL NO.	COLOUR OF SPOT	DISTANCE TRAVELLED BY THE COMPONE(cm)	DISTANCE TRAVELLED BY THE SOLVENT(cm)	R_f VALUE

$$R_f \text{ of } Pb^{2+} = \frac{\text{Distance moved by } Pb^{2+}}{\text{Distance moved by the solvent}} =$$

$$R_f \text{ of } Cd^{2+} = \frac{\text{Distance moved by } Cd^{2+}}{\text{Distance moved by the solvent}} =$$

EXPERIMENT NO:

DATE :

QUANTITATIVE ANALYSIS
REDOX TITRATIONS
PERMANGANOMETRIC TITRATIONS

TITRATION-1

AIM:
To prepare M/20 Mohrs salt solution [$FeS0_4.(NH_4)_2S0_4.6H_2O$] and to estimate the strength of the given $KMnO_4$ solution by titrating it against standard Mohrs salt solution

PRINCIPLE:
$KMnO_4$ acts as an oxidising agent. It oxidizes $Fe^{2+}$to Fe^{3+}ions and the oxidising agent gets reduced.

IONIC EQUATION:

$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

$5x\ [\ Fe^{2+} \longrightarrow Fe^{3+} + 1e^-]$

$MnO_{4-} + 5Fe^{2+} + 8H^+ \longrightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$

REQUIREMENTS:
Burette, pipette, conical flask,$KMnO_4$solution,Mohrs salt,4N H_2SO_4
$KMnO_4$ acts as a self indicator

END POINT:
Colourless turns pink

PROCEDURE:
1. Preparation of standard Mohr salt solution

(a) Weigh the given Mohrs salt
(b)Transfer the weighed Mohrs salt into 100ml standard flask through the funnel. Add 10ml of H_2SO_4.
(c) Make up the solution up to 100ml with H_2O.
(d)The last few drops may be added using a dropper.

2.Titration of Mohr's salt solution with $KMnO_4$ solution

(a)Wash the apparatus with water

(b)Rinse the burette with $KMnO_4$ solution and fill it up to the ‘0’ mark
(c) Rinse the pipette with the prepared Mohr's salt solution. Pipette out 20ml Mohr's salt solution into a clean flask.
(d)Add 4N H_2SO_4into it and titrate it against the $KMnO_4$ solution taken in a burette
(e)Repeat the experiment to get concordant value

OBSERVATION

ARTICLE WEIGHED	WEIGHT (g)
Weight of bottle + salt	
Weight of empty weighing bottle	
Weight of salt	

Mohr salt x $KMnO_4$

SL No	Volume of Mohr's salt (mL)	Burette (ml)		Volume of $KMnO_4$ (mL)
		Initial Reading	Final Reading	

Calculation

$$\frac{M1V1}{n1} = \frac{M2V2}{n2}$$

RESULT

Molarity of $KMnO_4$ solution= M

Strength of $KMnO_4$ solution= Molarity of $KMnO_4$ x 158

= x 158 = g/L

EXPERIMENT NO:

DATE:

VOLUMETRIC ANALYSIS
REDOX TITRATION
PERMANGANOMETRIC TITRATION

TITRATION - 2

AIM:

To prepare M/20 of oxalic acid to estimate the molarity and strength of given $KMnO_4$ solution.

PRINCIPLE:

IONIC EQUATION:

$[MnO_4^- + 8\ H^+ + 5\ e^- \rightarrow Mn^{2+} + 4\ H_2O] \times 2$

$[C_2O_4^{2-} \rightarrow 2\ CO_2 + 2\ e^-] \times 5$

$5\ C_2O_4^{2-} + 2\ MnO_4^- + 16\ H^+ \rightarrow 10\ CO_2 + 2\ Mn^{2+} + 8H$

REQUIREMENTS:

Burette, pipette, conical flasks, $KMnO_4$ solution, oxalic acid crystals and 4N H_2SO_4.

END POINT:

Colourless to pink.

INDICATOR : $KMnO_4$ act as a self indicator

PROCEDURE:

1. **PREPARATION OF STANDARD SOLUTION OF OXALIC ACID**

1. Weigh the given oxalic acid crystals.
2. Transfer the weighted oxalic acid crystals into 100ml standard flask through a funnel.
3. Now, make the solution up to the 100mL mark. The last few drops of water should be added using a dropper.

2. **TITRATION OF OXALIC ACID SOLUTION WITH $KMnO_4$ SOLUTION**

1. Wash the apparatus with water. Now rinse the burette with $KMnO_4$ solution. fill it with $KMnO_4$ solution up to the 0 mark. Rinse the pipette with oxalic acid solution. Pipette out 20ml of oxalic acid solution to a clean conical flask and heat it to 60-70°C, after adding 20ml of 4N H_2SO_4. Titrate it against the $KMnO_4$ solution taken in a burette.

2. End point is the appearance of pale pink colour. Repeat the experiment to get concordant values and record the observation.

OBSERVATION

ARTICLE WEIGHED	WEIGHT (g)
Weight of bottle + salt	
Weight of empty weighing bottle	
Weight of oxalic acid	

Oxalic acid x $KMnO_4$

SL No	Volume of oxalic acid (mL)	Burette (ml)		Volume of $KMnO_4$ (mL)
		Initial Reading	Final Reading	

CALCULATIONS:

$$\frac{M1V1}{n1} = \frac{M2V2}{n2}$$

Molarity of given $KMnO_4$ solution = M

Strength of given $KMnO_4$ solution = Molarity x Molecular mass of $KMnO_4$

= x 158

= g/L

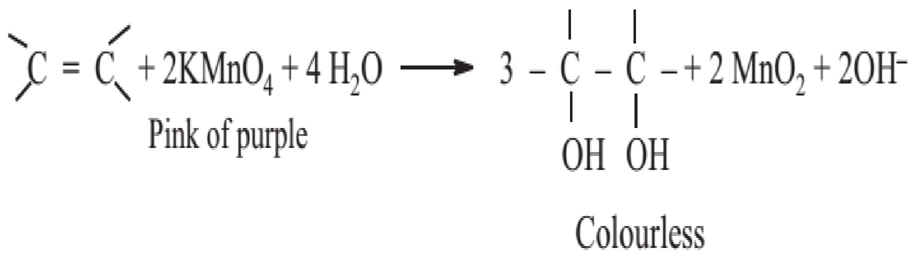
RESULT:

Molarity of given $KMnO_4$ solution = M

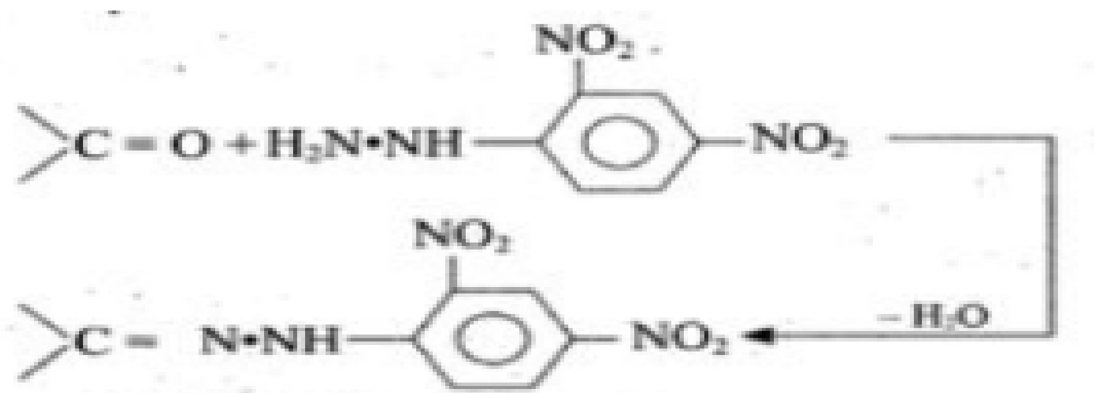
Strength of given $KMnO_4$ solution = g/L

ORGANIC ANALYSIS

GENERAL REACTIONS OF FUNCTIONAL GROUPS

1.TEST FOR UNSATURATION		
To a little of the organic compound Baeyer's reagent is added.	Pink colour is discharged.	Presence of unsaturation.
		
2. TEST FOR ALCOHOLS		
To a little of the organic compound in a test tube a small piece of metallic sodium is added.	Brisk effervescence due to liberation of hydrogen gas.	Presence of hydroxyl group.
$2ROH + 2Na \longrightarrow 2RONa + H_2$		
Esterification: To a little of the organic compound acetic acid is added followed by 2 drops of conc. H_2SO_4. Warm the	A fruity smell due to the formation of ester.	Presence of hydroxyl group.
contents in a water bath and pour it into cold water.		

$CH_3COOH + ROH \rightarrow CH_3COOR + H_2O$		
3.<u>TEST FOR CARBOXYLIC ACID</u>		
To a little of the organic compound saturated solution of sodium bicarbonate is added.	Effervescence due to formation of CO_2. $RCOOH + NaHCO_3 \rightarrow RCOONa + H_2O + CO_2$	Presence of carboxylic acid.
Esterification: To a little of the organic compound Ethyl alcohol and 2 drops of conc.H_2SO_4 are added. Warm the contents in a water bath and pour it into cold water.	A fruity smell due to formation of ester. $RCOOH + C_2H_5OH \rightarrow RCOOC_2H_5 + H_2O$	Presence of carboxylic acid.
4.<u>TEST FOR PHENOLS</u>		
To a little of the organic compound neutral ferric chloride is added and is diluted with water.	Violet colourisation is developed.	Presence of phenols.
$6C_6H_5OH + FeCl_3 \rightarrow [(C_6H_5O)_3Fe] + 3HCl + 3H^+$		

<u>Liebermann's Nitroso reaction:</u> Take a little of the sample, add $NaNO_2$ followed by concentrated H_2SO_4. Then dilute with water. Add NaOH.	Appearance of a deep green or blue colour which changes to red on dilution with water and restores original green or blue colour with NaOH.	Presence of phenols.
5.<u>TEST FOR CARBONYL COMPOUNDS</u>		
To a little of the organic compound 2,4-dinitro phenyl hydrazine reagent is added.	Orange yellow crystals appear due to the formation of phenyl hydrazone.	Presence of carbonyl compounds.
 2, 4 Dinitro phenyl hydrazones (Orange or yellow ppt.)		
<u>Silver Mirror/Tollen's Reagent</u> **<u>Test:</u>**To a little of the organic compound Tollen's reagent(Ammoniacal silver nitrate) is added, warmed over water bath for 5 minutes.	Appearance of silver lining along the sides of the test tube.	Presence of aldehyde group.
$RCHO + 2[Ag(NH_3)_2]^{2+} + 3OH^- \rightarrow 2Ag + 2H_2O + 4NH_3 + RCOO^-$		

Fehling's Solution Test: To a little of the organic compound equal amounts of Fehling's reagent A and B are added and warmed in water bath.	Red precipitate is formed. $RCHO + Cu^{2+} + 5OH^{-} \rightarrow RCOO^{-} + Cu_2O$ $3H_2O$	Presence of aldehyde group.

Sodium Nitroprusside Test: Add a small amount of organic compound to Sodium nitroprusside solution followed by NaOH solution in drops.	Red colourisation.	Presence of ketones.
$CH_3COCH_3 + OH^{-} \rightarrow CH_3COCH_2^{-} + H_2O$ $[Fe(CN)_5NO]^{2-} + CH_3COCH_2^{-} \rightarrow [Fe(CN)_5NOCH_3COCH_2]^{3-}$		
6.TEST FOR AMINES		
To a little of the organic compound dil.HCl is added and shaken well.	Organic compound is soluble.	Presence of amines.
$RNH_2 + HCl \rightarrow RNH_3Cl^{-}$		
Dye Test: Dil. HCl is added to sodium nitrite crystals in a test tube. This solution is then treated with organic compound in another test tube. B-naphthol is dissolved in sodium hydroxide solution. The two solutions are mixed together.	Orange red colouration.	Presence of amines.